

Data Card

Input vs. Output

Input: A farmer's natural-language question (e.g., "Why are my maize leaves turning yellow?"). Output: A ranked list of the top 5 most relevant document IDs from a knowledge base of 695 agricultural extension factsheets, ordered from most to least relevant.

Classification Problem

Not a classification task in the traditional sense. This is a ranking & retrieval task. Rather than predicting discrete labels, the model scores and ranks candidate documents by relevance to a query. Relevance itself is graded on a 0–3 scale (0 = not relevant, 1 = marginal/different crop, 2 = related aspect, 3 = perfect answer), which functions similarly to an ordinal classification of relevance per document.

Labeling

Relevance labels (qrels) were provided by the competition organizers, derived from expert judgment on which documents genuinely answer each training query. We did not create these labels ourselves & we rely on the pre-supplied qrels_train.csv for local evaluation and trust the organizer's hidden expert judgments for the true evaluation set. We check consistency by using the provided evaluate_ndcg_at_5 function to validate our own retrieval outputs against the known training labels before submitting.

Data Creation

The document corpus (695 factsheets) and query sets were provided directly by the competition organizers, we are not collecting or scraping this data ourselves. Per the brief, the corpus mixes original factsheets with rewrites of real CC-BY-licensed agricultural research, meaning it is derived from legitimately licensed real-world agricultural knowledge rather than synthetic or scraped-without-permission content.

Web Collection Ethics

We are not independently sourcing data from the web for this project. The underlying corpus was already curated by the organizers from CC-BY-licensed agricultural research, which permits reuse with attribution. We are not introducing new web-scraped data, so no additional consent, copyright, or scraping-ethics concerns arise from our side of the work.

Bias and Representation

The dataset is described as covering crop diseases, pests, nutrient deficiencies, soil management, climate adaptation, and fertilizer advice for Sub-Saharan African agriculture but we do not know the exact crop/regional distribution across the 695 factsheets. Crops or regions with less documented research e.g., minor/local crop varieties, or regions with fewer formal extension resources may be underrepresented, meaning our retrieval system

could perform less reliably for farmers growing those crops. We plan to check this by reviewing which crops & topics appear in our lowest-nDCG@5 training queries, to see if certain crops are systematically harder to retrieve for.

Values and Community Alignment

Our dataset design supports the values identified in Challenge 1 i.e. accuracy, accessibility, and trustworthiness, because working with a real, expert-labeled dataset, rather than one we scraped or synthesized ourselves, ensures our evaluation reflects genuine expert judgment about what farmers actually need. We remain accountable to the intended community by being transparent about where the dataset's coverage may be uneven, and by prioritizing retrieval quality on hard negatives i.e. documents that look similar but are wrong, rather than optimizing only for leaderboard performance.